Data Cleaning & Validation Report

Week 2 – Data Cleaning, Integration & Validation

Data Visualization Associate Internship — Excelerate
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Datasets: Opportunity Data | Learner Data

1. Data Integration Process

1.1 Objective

The objective of the data integration process was to combine the cleaned **Opportunity** and **Learner** datasets into a single integrated dataset to support comprehensive analysis and dashboard development. The integration enables learner application information to be analyzed alongside opportunity details within a unified table.

1.2 Datasets Used

Dataset	Description	Final Records
Opportunity (`opportunity_clean`)	Contains information about opportunities, including category, duration, fee, location, eligibility, and rewards.	**5,728**
Learner (`learner_clean`)	Contains learner application records, payment status, workflow status, cohort details, and application history.	**14,793**

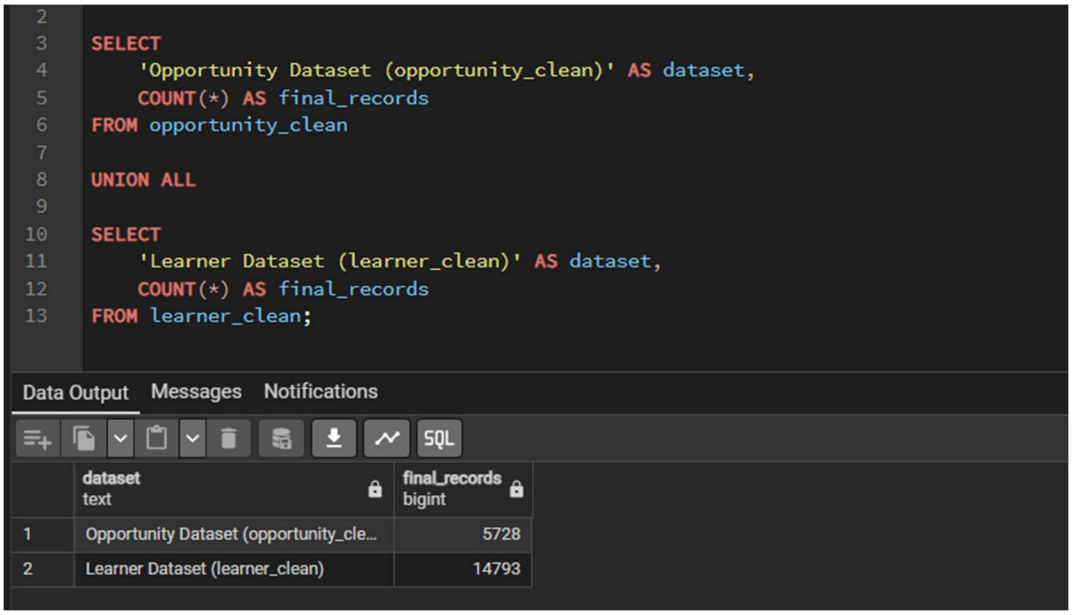


Fig 1: Total Rows.

Opportunity Dataset (opportunity_clean)	Learner Dataset (learner_clean)
Purpose: Stores details of available opportunities offered on the platform.	**Purpose:** Stores learner profiles and their application history.
Final Records: 5,728	**Final Records:** 14,793
Primary Identifier: opportunity_id	**Primary Identifier:** sk (Learner ID)
Key Attributes: Opportunity title, category, duration, fee/payment status, location, eligibility criteria, rewards, application dates, and related metadata.	**Key Attributes:** Learner ID, payment status, workflow status, cohort details, application history, learner category, and associated opportunity reference.
Cleaning Performed: Removed corrupted rows, standardized location values, cleaned category values, validated timestamps, converted invalid values to NULL, and removed duplicates.	**Cleaning Performed:** Removed corrupted/Base64 records, standardized category values, validated payment and workflow status, removed invalid records, and retained consistent text datatypes.
Role in Integration: Provides master information about each opportunity.	**Role in Integration:** Provides learner participation and application information linked to opportunities.

1.3 Join Logic Used

Initially, no direct common key appeared to exist between the two datasets. A detailed examination of the identifier columns revealed that the learner table's **sk** field and the opportunity table's **opportunity_id** field contained the same underlying unique identifier but used different prefixes.

Learner Dataset	Opportunity Dataset
`Learner#0000000010Z56C78JF7M6VQXQ2`	`Opportunity#0000000010Z56C78JF7M6VQXQ2`

To establish the relationship, the prefixes (Learner# and Opportunity#) were removed using PostgreSQL's REPLACE() function, allowing both datasets to be joined on the common identifier.

```
CREATE TABLE learner_opportunity AS
SELECT
   l.*,

   o.opportunity_id,
   o.badge,
   o.careeraddon,
   o.code,
   o.currency_type,
   o.current_editor,
   o.dropouttransaction,
   o.duration,
   o.duration_type,
   o.eligibility,
   o.image_link,
   o.is_archived,
   o.is_auto_approve,
   o.last_date_to_apply,
   o.long_description,
   o.microscholarship,
   o.name AS opportunity_name,
   o.notstartedtransaction,
   o.panellist,
   o.role,
   o.role_responsibility,
   o.short_description,
   o.summary,
   o.testimonial,
   o.tracking_questions,

   o.pk AS opportunity_pk,
   o.category AS opportunity_category,
   o.fee AS opportunity_fee,
   o.location AS opportunity_location,
   o.reward AS opportunity_reward,
   o.created_at AS opportunity_created_at,
   o.modified_at AS opportunity_modified_at

FROM learner_clean l
JOIN opportunity_clean o
ON REPLACE(l.sk, 'Learner#', '') =
  REPLACE(o.opportunity_id, 'Opportunity#', '');
```

An INNER JOIN was used to retain only records that had matching identifiers in both datasets

1.4 Key Fields Involved

Table	Key Field	Purpose
`learner_clean`	`sk`	Unique learner identifier used for integration
`opportunity_clean`	`opportunity_id`	Unique opportunity identifier used for integration

Additional opportunity attributes such as category, duration, fee, location, eligibility, and rewards were merged with learner application information to create a comprehensive analytical dataset.

1.5 Structure of the Integrated Dataset

A new table named **learner_opportunity** was created to store the integrated data.

The integrated dataset contains:

- Learner application details
- Opportunity information
- Payment status
- Workflow status
- Cohort information
- Opportunity category
- Opportunity location
- Fee details
- Eligibility information
- Reward details
- Application and opportunity timestamps
- ……

To prevent naming conflicts, duplicate column names such as pk, category, fee, created_at, and modified_at were renamed during table creation (for example, opportunity_category, opportunity_fee, and opportunity_created_at).

The integration process successfully produced **14,791** matched records, confirming that the relationship between the two datasets was established correctly and that the resulting dataset is suitable for further validation and analytical reporting.

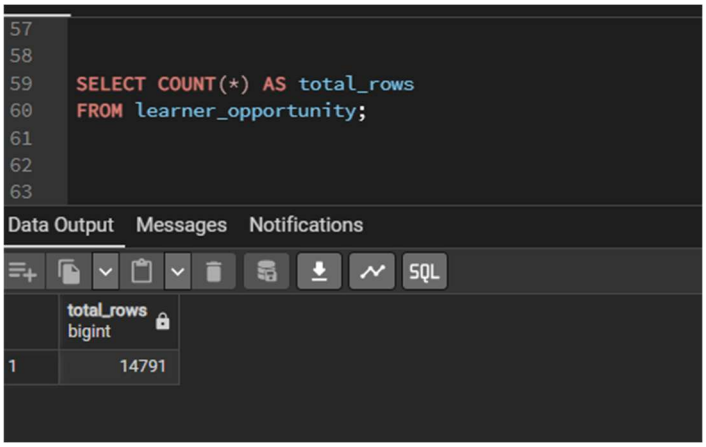


Fig 2: Joined table Total Rows.

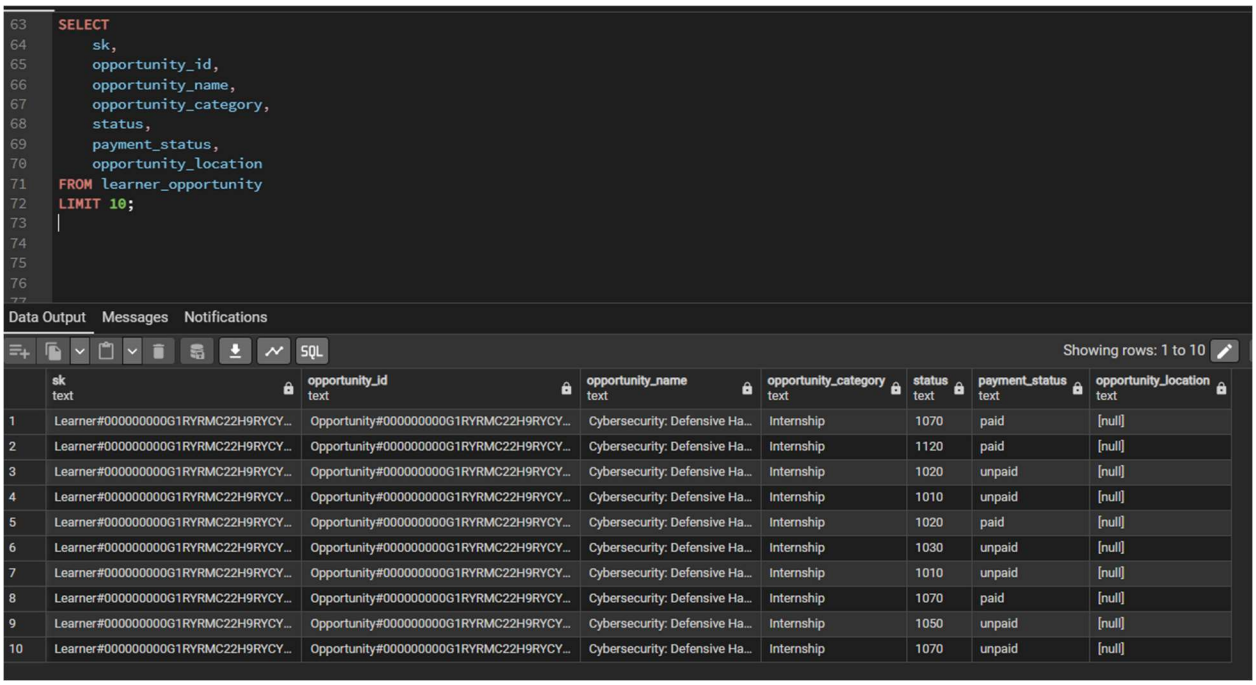


Fig 3: Learner_Oppurtunity Table

2. Data Cleaning Activities

2.1 Objective

The objective of the data cleaning process was to improve the quality, consistency, and reliability of the Opportunity and Learner datasets before performing analytical reporting and dashboard development. Data cleaning involved identifying corrupted records, standardizing inconsistent values, validating key attributes, and preparing the integrated dataset for analysis.

2.2 Cleaning Steps Performed

Opportunity Dataset (opportunity_clean)

The Opportunity dataset was cleaned to improve data quality, consistency, and reliability before integration with the learner dataset. The following cleaning activities were performed:

- Imported the raw Opportunity dataset containing 5,733 records into PostgreSQL.
- Identified and removed 5 corrupted records containing invalid `opportunity_id` values, malformed JSON fragments, and inconsistent testimonial data that could affect downstream analysis.
- Verified the uniqueness of the `opportunity_id` field and confirmed that no duplicate opportunity records remained after cleaning.
- Standardized inconsistent location values by converting variations such as *virtual*, *Virtual*, *WORK FROM HOME*, *WFM*, *wfm*, and *vitrual* into consistent formats (`Virtual` and `Work From Home`).
- Cleaned the category column by replacing invalid values such as "XploreU" with SQL `NULL` to represent missing or invalid data consistently.
- Converted textual "NULL" strings across relevant columns into SQL `NULL` values to ensure proper handling of missing data during analysis.
- Reviewed and validated date-related fields to identify malformed or inconsistent timestamp values and retained only valid records.
- Verified that the `pk` column contains the constant identifier "Opportunity#" and serves as a prefix rather than a unique primary key.
- Performed final validation to confirm data consistency, record counts, and overall dataset integrity after cleaning.

Final Opportunity Dataset Summary:

- Original Records: 5,733
- Corrupted Records Removed: 5
- Final Clean Records: 5,728

```
SELECT DISTINCT location
FROM opportunity_clean
ORDER BY location;
```

```sql
SELECT DISTINCT location
FROM opportunity_clean
ORDER BY location;
```

Data Output | Messages | Notifications

	location text
1	Aliquam officia est
2	Animi iste aut anim
3	Commodi et aut sunt
4	Elit cum quis volup
5	kolkata
6	Location
7	Opportunity name Page Testi...
8	Qui voluptatibus nat
9	Quia ad ipsum quos
10	Quis voluptate totam
11	qwerty
12	Recusandae Ex volup
13	Virtual
14	Work From Home
15	[null]

Fig 4: Standardized opportunity location values after cleaning

```sql
SELECT DISTINCT category
FROM opportunity_clean
ORDER BY category;
```

Data Output | Messages | Notifications

	category text
1	Career
2	Competition
3	Course
4	Engagement
5	Event
6	Internship
7	JobSimulati...
8	Masterclass
9	Program
10	[null]

Fig 5: Opportunity categories after cleaning and standardization

Learner Dataset (learner_clean)

The following cleaning activities were performed:

- Imported the raw Learner dataset containing 15,029 records.
- Removed 236 corrupted records containing embedded PDF/Base64 data.
- Cleaned invalid category values by replacing "XploreU" with SQL NULL.
- Validated workflow status values without modifying legitimate source values.
- Validated payment_status values (paid, unpaid).
- Reviewed date-related columns for consistency.
- Performed missing value analysis on important fields.

Final Learner records: 14,793

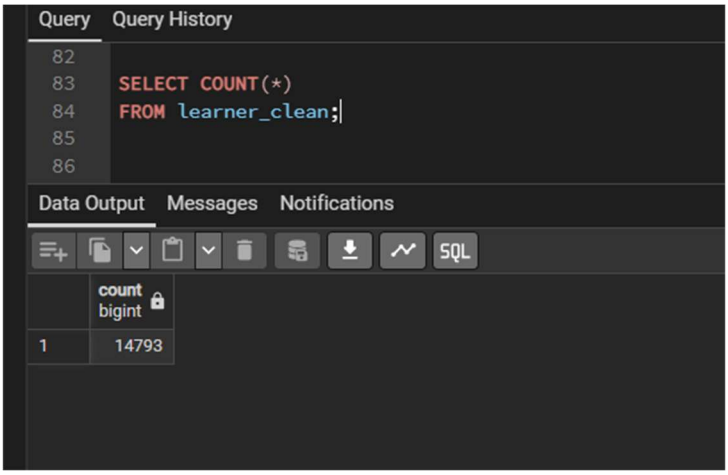


Fig 6: Learner dataset after removing corrupted records

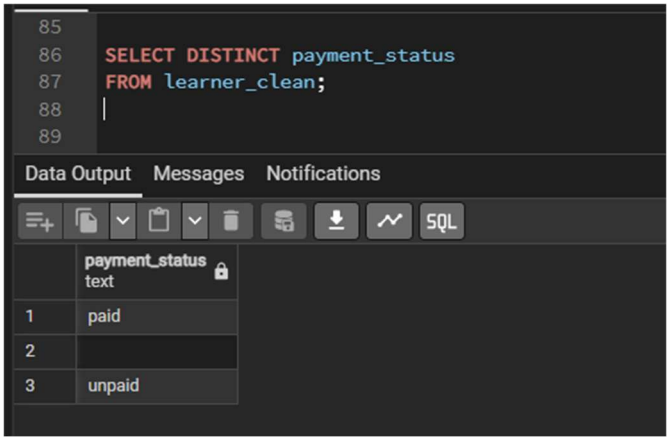


Fig 7: Valid payment status values in the learner dataset

Integrated Dataset (learner_opportunity)

After integrating both datasets, additional cleaning was performed:

- Renamed duplicate column names (such as pk, category, fee, created_at, and modified_at) to avoid naming conflicts.
- Converted blank status values ('') to SQL NULL.
- Converted blank opportunity_location values ('') to SQL NULL.
- Investigated missing values and validated that they originated from the source datasets rather than the integration process.

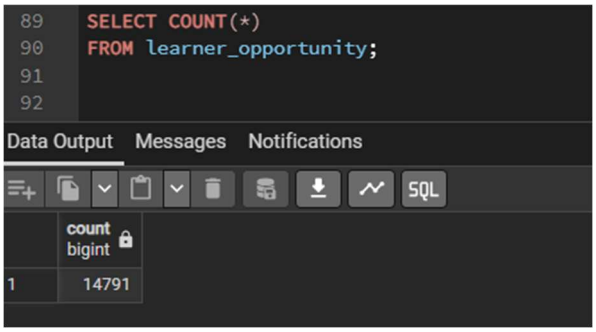


Fig 8: Integrated dataset created successfully with 14,791 matched records.

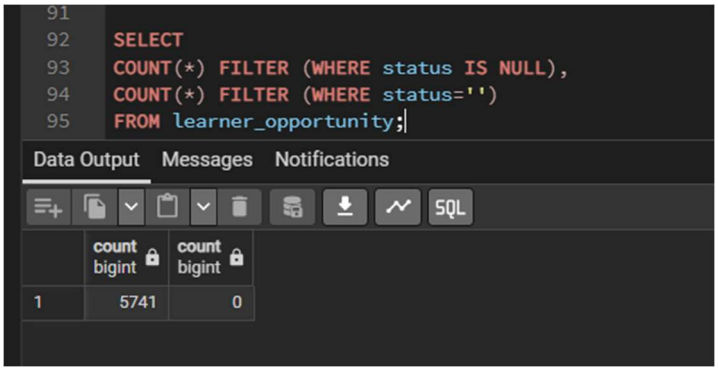


Fig 9: Verification that blank workflow status values were standardized to SQL NULL.

2.3 Assumptions Made

The following assumptions were made during data cleaning:

- The identifier contained in learner_clean.sk and opportunity_clean.opportunity_id represents the same underlying entity after removing the prefixes Learner# and Opportunity#.
- Invalid category values (XploreU) were treated as missing information and converted to SQL NULL.
- Blank strings ('') represent missing values and were standardized to SQL NULL.
- Missing opportunity names and locations were retained as NULL because no reliable source existed to infer or replace them.
- Workflow status values 103 and 110 were retained because there was no evidence that they were incorrect or truncated.

2.4 Rationale Behind Major Decisions

The following decisions were taken to maintain data integrity:

- Corrupted records were removed because they could affect query results and downstream analysis.
- Inconsistent categorical values were standardized to ensure accurate grouping and aggregation.
- Textual "NULL" values and blank strings were converted to SQL NULL to maintain consistency in missing value handling.
- Duplicate column names were renamed during integration to avoid ambiguity in SQL queries.
- Source-data anomalies, such as missing opportunity names and uncommon workflow status codes, were preserved instead of being modified, ensuring that the cleaned dataset accurately reflects the original data without introducing assumptions.

3. Validation & Comparison

3.1 Objective

After completing the data cleaning and integration processes, the datasets were validated to ensure data quality, consistency, and integrity. Validation was performed by comparing the datasets before and after cleaning, verifying key identifiers, checking for missing values, and confirming that the integrated dataset was suitable for analysis and dashboard development.

3.2 Before vs. After Cleaning

Opportunity Dataset

Validation Metric	Before Cleaning	After Cleaning
Total Records	5,733	**5,728**
Corrupted Records	Present	**Removed**
Duplicate Opportunity IDs	Checked	**0 Duplicates**
Inconsistent Location Values	Present	**Standardized**
Invalid Categories	Present (`XploreU`)	**Converted to NULL**
Invalid "NULL" Strings	Present	**Converted to SQL NULL**

Learner Dataset

Validation Metric	Before Cleaning	After Cleaning
Total Records	15,029	**14,793**
Corrupted PDF/Base64 Rows	Present	**Removed**
Invalid Categories	Present	**Converted to NULL**
Payment Status	Mixed	**Validated**
Workflow Status	Reviewed	**Validated**

Integrated Dataset

Validation Metric	Result	Validation Metric
Matched Records	**14,791**	Matched Records
Join Validation	Successful	Join Validation
Duplicate Opportunity IDs	**0**	Duplicate Opportunity IDs
Duplicate Column Names	Resolved	Duplicate Column Names
Blank Status Values	Converted to SQL NULL	Blank Status Values
Blank Location Values	Converted to SQL NULL	Blank Location Values

3.3 Improvements Made

The following improvements were achieved during the cleaning process:

- Removed corrupted records from both datasets.
- Standardized inconsistent categorical values.
- Normalized location values into consistent formats.
- Converted textual "NULL" values and blank strings into SQL NULL.
- Renamed duplicate columns during integration to avoid ambiguity.
- Successfully integrated the datasets using normalized identifiers.
- Preserved valid source-data anomalies without introducing artificial values.

3.4 Changes in Data Quality

The overall quality of the datasets improved significantly after cleaning and validation.

Key improvements include:

- Reduced the number of corrupted records.
- Improved consistency of categorical attributes.
- Eliminated duplicate opportunity identifiers.
- Standardized missing value representation.
- Verified the integrity of the integrated dataset.

- Confirmed that all analytical queries execute successfully on the cleaned data.

3.5 Validation Checks Performed

The following validation checks were carried out after integration:

Record Count Validation

- Opportunity Dataset: 5,728 records
- Learner Dataset: 14,793 records
- Integrated Dataset: 14,791 matched records

Duplicate Validation

Verified that the Opportunity dataset contains no duplicate opportunity_id values.

Missing Value Validation

Final validation results:

Column	Missing Values
Opportunity Name	**1**
Opportunity Category	**7**
Payment Status	**0**
Status	**5,741**
Opportunity Location	**3,718**

The remaining missing values represent source-data characteristics or intentionally standardized NULL values and were retained to preserve data integrity.

Workflow Status Validation

Workflow status values were reviewed to identify unexpected codes. The uncommon values (103 and 110) were investigated and retained because there was no evidence that they were incorrect or required modification.

Join Validation

The integration between the learner and opportunity datasets was verified by matching normalized identifiers after removing the Learner# and Opportunity# prefixes. The validation confirmed **14,791** successfully matched records.

```sql
SELECT 'Opportunity Raw' AS dataset, COUNT(*) AS total_records
FROM opportunity_raw

UNION ALL

SELECT 'Opportunity Clean', COUNT(*)
FROM opportunity_clean

UNION ALL

SELECT 'Learner Raw', COUNT(*)
FROM learner_raw

UNION ALL

SELECT 'Learner Clean', COUNT(*)
FROM learner_clean;
```

Data Output | Messages | Notifications

	dataset text	total_records bigint
1	Learner Raw	15029
2	Learner Clean	14793
3	Opportunity Raw	5733
4	Opportunity Cle...	5728

Fig 10: Table Before and After

opportunity_raw	opportunity_clean	learner_raw	learner_clean
5733	5728	15029	14793

```sql
SELECT
    opportunity_id,
    COUNT(*)
FROM opportunity_clean
GROUP BY opportunity_id
HAVING COUNT(*) > 1;
```

Data Output | Messages | Notifications

	opportunity_id text	count bigint

Fig 11: Duplicate Opportunity ID Validation

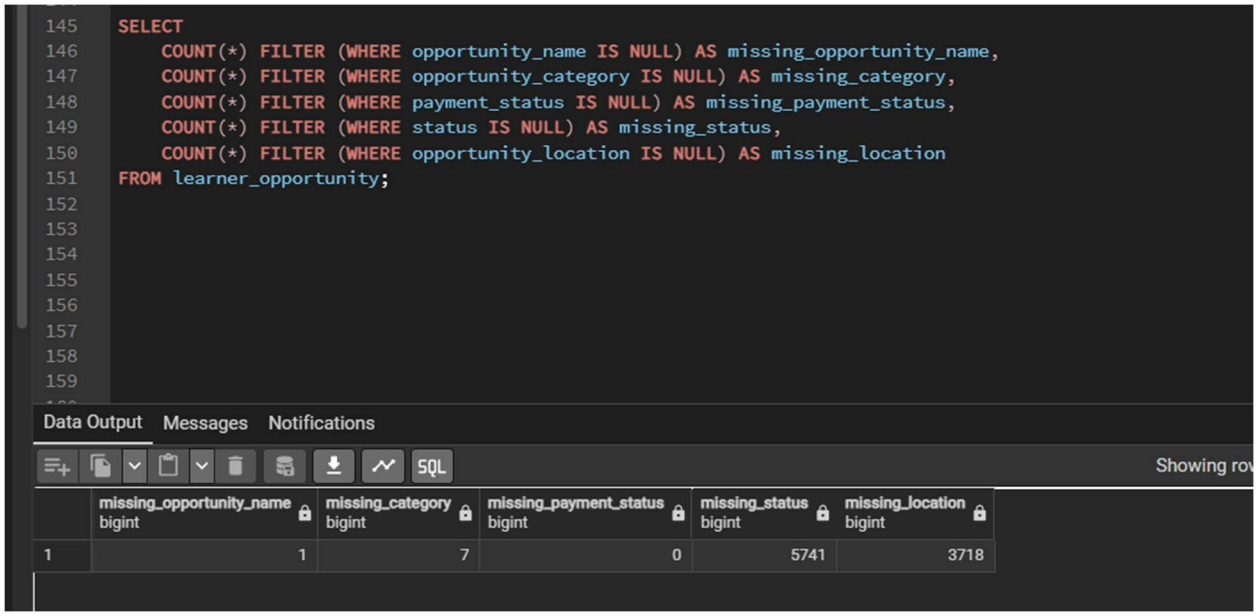


Fig 12: Missing Value Validation

Name	Category	Payment	Status	Location
1	7	0	5741	3718

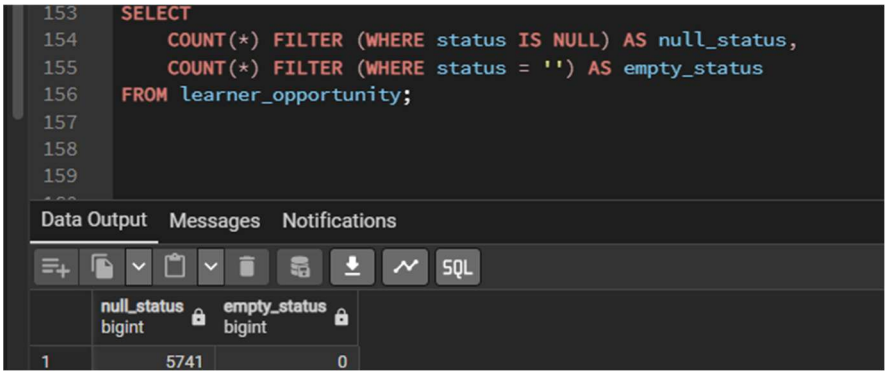


Fig 13: Blank Values Converted to NULL

4. Updated Analysis

4.1 Objective

After completing the data integration, cleaning, and validation processes, exploratory SQL analysis was performed on the integrated dataset (learner_opportunity). The objective was to verify that the dataset is ready for reporting and dashboard development by generating key performance indicators (KPIs), summary statistics, and analytical insights.

4.2 Key Performance Indicators (KPIs)

The following SQL query was executed to generate the primary dashboard KPIs.

```
SELECT
    COUNT(*) AS total_applications,
    COUNT(DISTINCT sk) AS unique_learners,
    COUNT(DISTINCT opportunity_id) AS unique_opportunities
FROM learner_opportunity;
```

Analysis:

- Total Applications
- Unique Learners
- Unique Opportunities

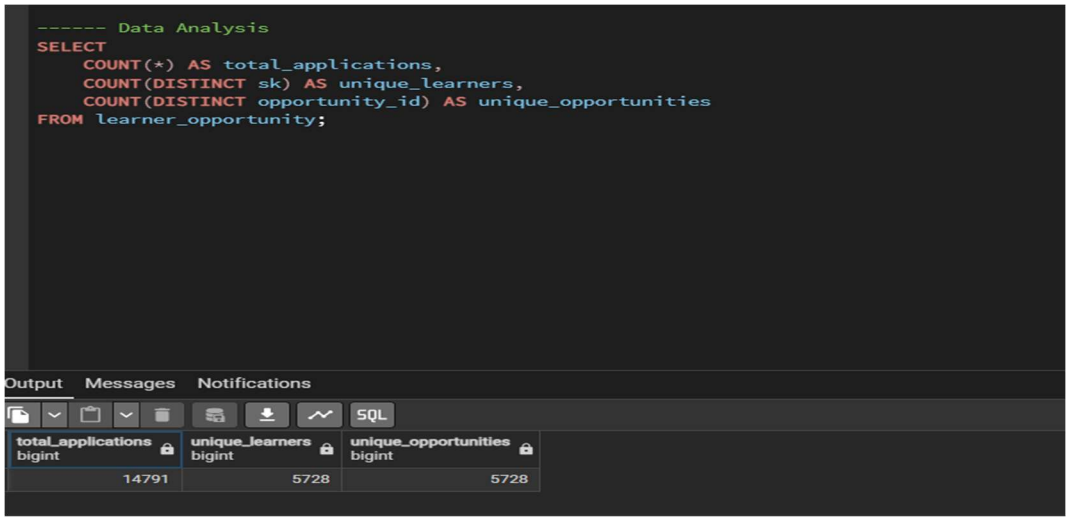


Fig 14: KPI Summary

4.3 Opportunity Category Analysis

Applications were grouped by opportunity category to identify the most popular opportunity types.

```
SELECT
    opportunity_category,
    COUNT(*) AS applications
FROM learner_opportunity
GROUP BY opportunity_category
ORDER BY applications DESC;
```

Analysis:

- Identifies the categories receiving the highest number of applications.
- Useful for category-wise dashboard visualizations.

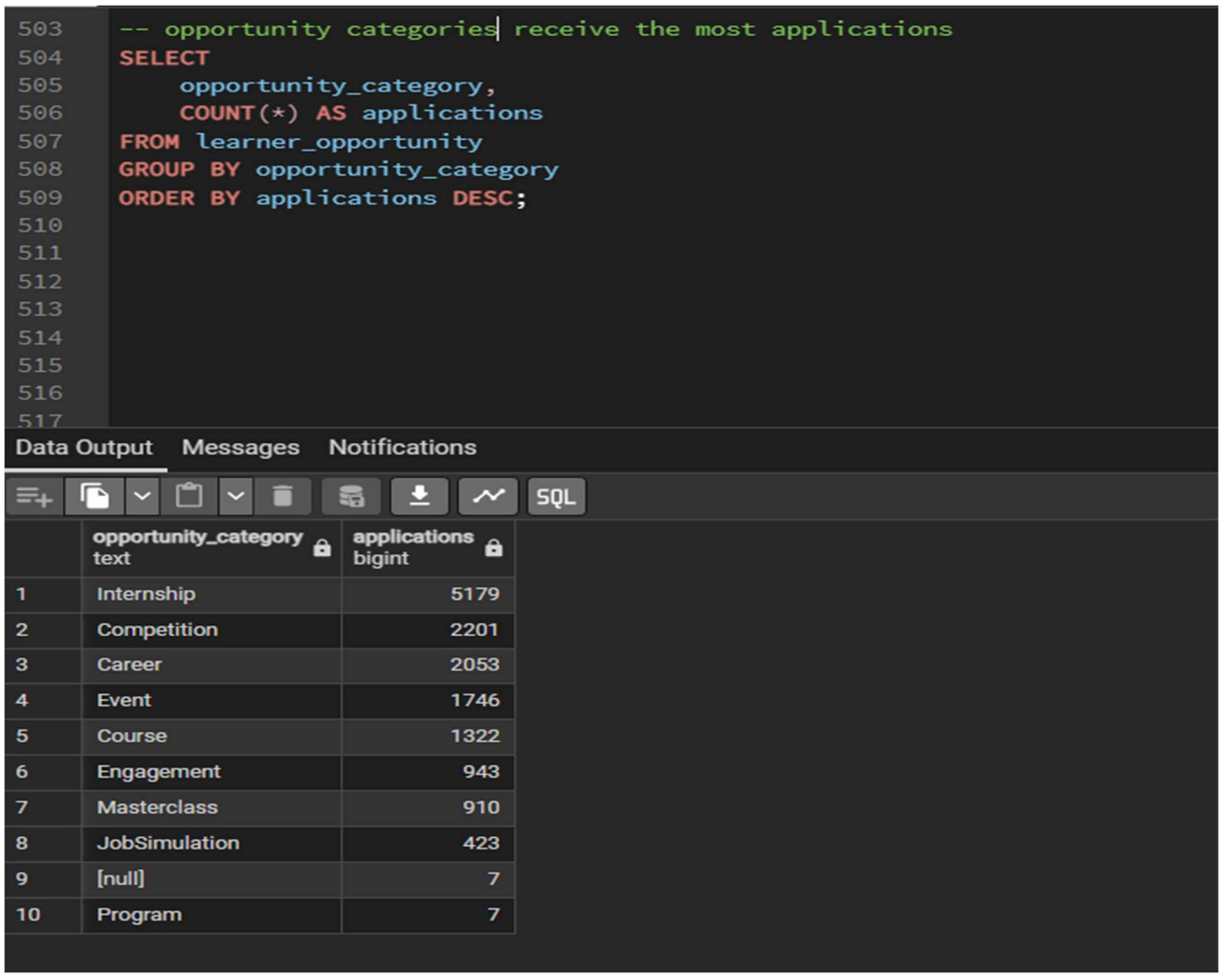


Fig 16: Payment Status Distribution

4.4 Payment Status Analysis

The distribution of paid and unpaid applications was analyzed.

```
SELECT
    payment_status,
    COUNT(*) AS total
FROM learner_opportunity
GROUP BY payment_status
ORDER BY total DESC;
```

Analysis:

- Compares paid versus unpaid applications.
- Useful for financial reporting.

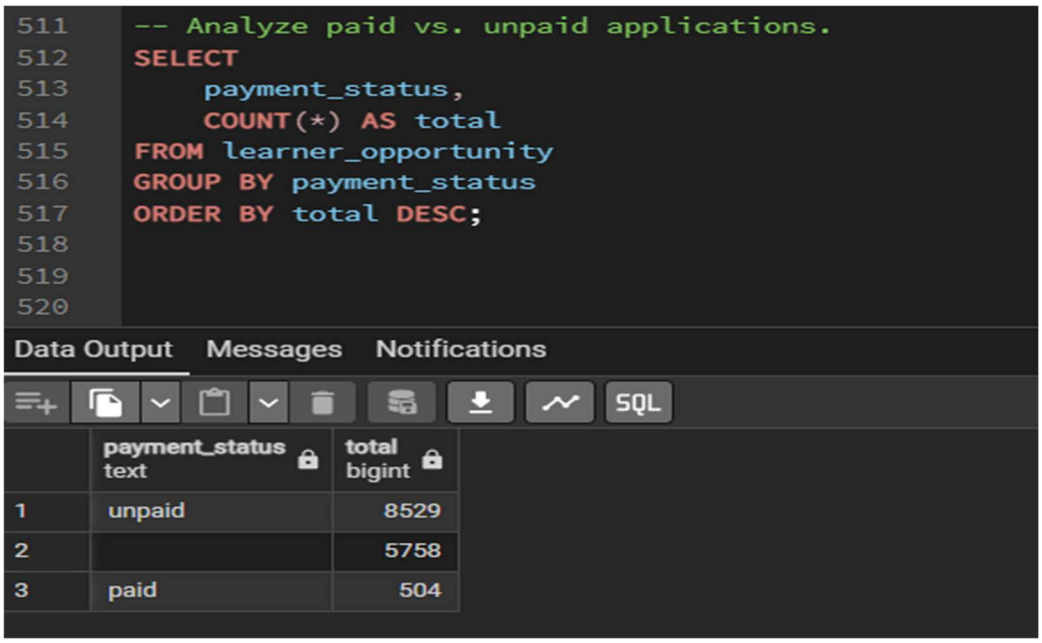


Fig 17: Workflow Status Distribution

4.5 Workflow Status Analysis

Application workflow stages were summarized using the workflow status field.

```
SELECT
    status,
    COUNT(*) AS total
FROM learner_opportunity
```

GROUP BY status

ORDER BY total DESC;

Analysis:

- Displays the distribution of applications across workflow stages.
- Helps monitor application progress.

```
518
519  -- Understand application workflow stages
520  SELECT
521      status,
522      COUNT(*) AS total
523  FROM learner_opportunity
524  GROUP BY status
525  ORDER BY total DESC;
526
```

Data Output | Messages | Notifications

SQL

	status text	total bigint
1		5741
2	1070	3335
3	1010	1791
4	1120	1123
5	1030	694
6	1020	597
7	1110	435
8	1055	348
9	1050	326
10	1040	196
11	1011	101
12	1080	94
13	1100	4
14	1090	2
15	103	2
16	110	1
17	1060	1

Fig 18: Workflow Status Distribution

4.6 Top Opportunities

The most frequently applied opportunities were identified.

```
SELECT
    opportunity_name,
    COUNT(*) AS applications
FROM learner_opportunity
GROUP BY opportunity_name
ORDER BY applications DESC
LIMIT 10;
```

Analysis:

- Identifies the most popular opportunities.
- Useful for ranking and performance analysis.

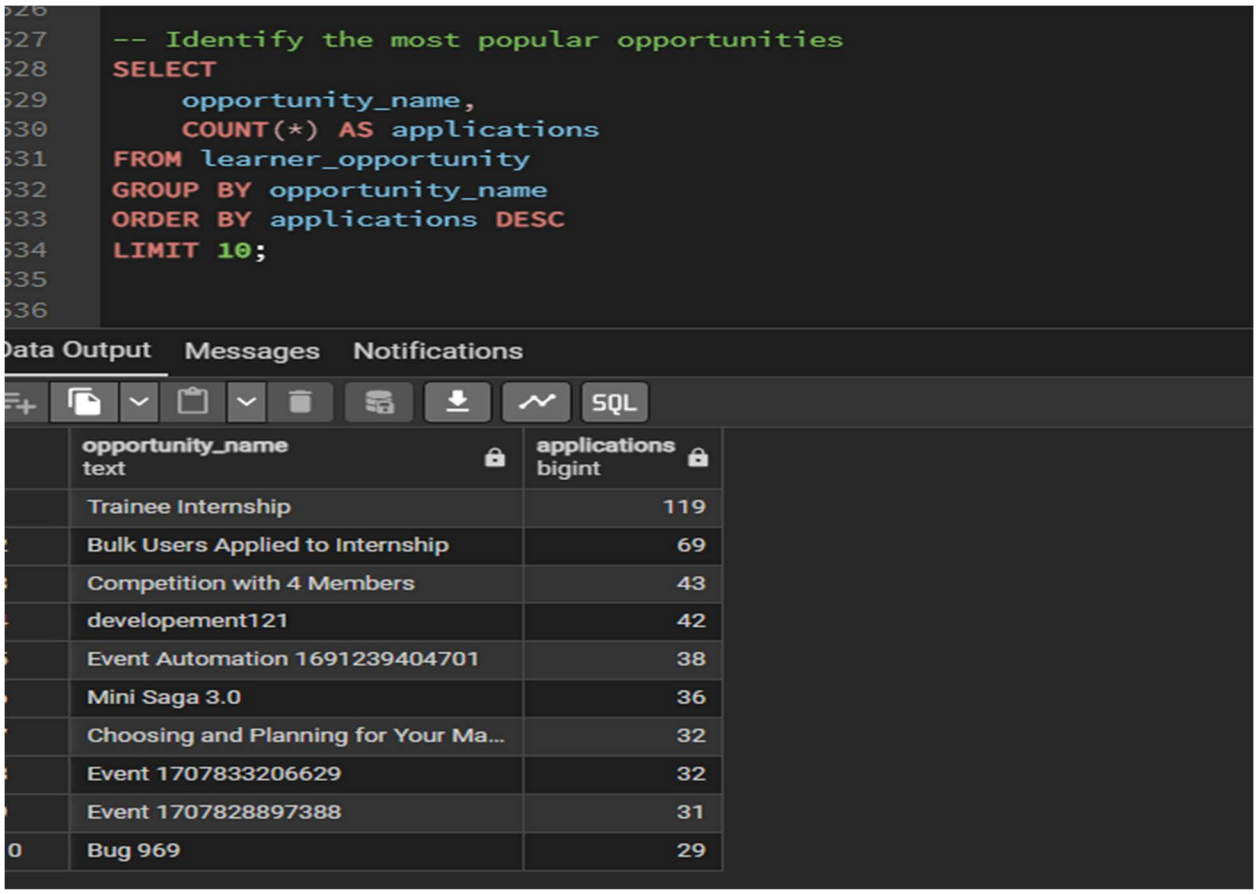


Fig 19: Top 10 Most Applied Opportunities

4.7 Opportunity Location Analysis

Application distribution across opportunity locations was analyzed.

SELECT

opportunity_location,

COUNT(*) AS applications

FROM learner_opportunity

GROUP BY opportunity_location

ORDER BY applications DESC;

Analysis:

- Shows the preferred opportunity locations.
- Useful for geographical or work-mode insights.

```
536  -- preferred work locations
537  SELECT
538      opportunity_location,
539      COUNT(*) AS applications
540  FROM learner_opportunity
541  GROUP BY opportunity_location
542  ORDER BY applications DESC;
543
544
545
```

Data Output | Messages | Notifications

SQL

	opportunity_location text	applications bigint
1	Work From Home	6076
2	Virtual	4941
3	[null]	3718
4	Location	14
5	Quia ad ipsum quos	11
6	Commodi et aut sunt	10
7	Opportunity name Page Testi...	7
8	Quis voluptate totam	5
9	Recusandae Ex volup	2
10	Qui voluptatibus nat	2
11	Elit cum quis volup	1
12	kolkata	1
13	Aliquam officia est	1
14	Animi iste aut anim	1
15	qwerty	1

Fig 20: Applications by Opportunity Location

4.8 Opportunity Fee Analysis

Applications were analyzed based on the opportunity fee.

```
SELECT
    opportunity_fee,
    COUNT(*) AS total
FROM learner_opportunity
GROUP BY opportunity_fee
ORDER BY total DESC
LIMIT 15;
```

Analysis:

- Shows application trends across different fee amounts.
- Useful for evaluating participation based on fee structure.

```
544
545  -- Understand application distribution by fee
546  SELECT
547      opportunity_fee,
548      COUNT(*) AS total
549  FROM learner_opportunity
550  GROUP BY opportunity_fee
551  ORDER BY total DESC
552  LIMIT 15;
553
554
```

Data Output | Messages | Notifications

SQL

	opportunity_fee text	total bigint
1	0	11708
2	1	1413
3	100	539
4	10	363
5	2	61
6	500	48
7	50	39
8	5	30
9	23	29
10	14	27
11	2000	27
12	15	23
13	12	22
14	20	21
15	111	16

Fig 21: Opportunity Fee Distribution

4.9 Duration Type Analysis

The duration of opportunities was summarized.

```
SELECT
    duration_type,
    COUNT(*) AS total
FROM learner_opportunity
GROUP BY duration_type
ORDER BY total DESC;
```

Analysis:

- Identifies the most common opportunity durations.

```
555  -- the most common opportunity durations
556  SELECT
557      duration_type,
558      COUNT(*) AS total
559  FROM learner_opportunity
560  GROUP BY duration_type
561  ORDER BY total DESC;
562
563
```

Data Output | Messages | Notifications

SQL

	duration_type text	total bigint
1	weeks	9650
2	month	1151
3	hours	1007
4	days	956
5	months	679
6	week	432
7	minutes	329
8	years	209
9	day	168
10	hour	110
11	minute	58
12	year	28
13	da	8
14	yearssss	4
15	NULL	2

Fig 22: Opportunity Duration Analysis

4.10 Auto Approval Analysis

The proportion of auto-approved opportunities was analyzed.

```
SELECT
   is_auto_approve,
   COUNT(*) AS total
FROM learner_opportunity
GROUP BY is_auto_approve;
```

Analysis:

- Compares automatically approved and manually reviewed opportunities.

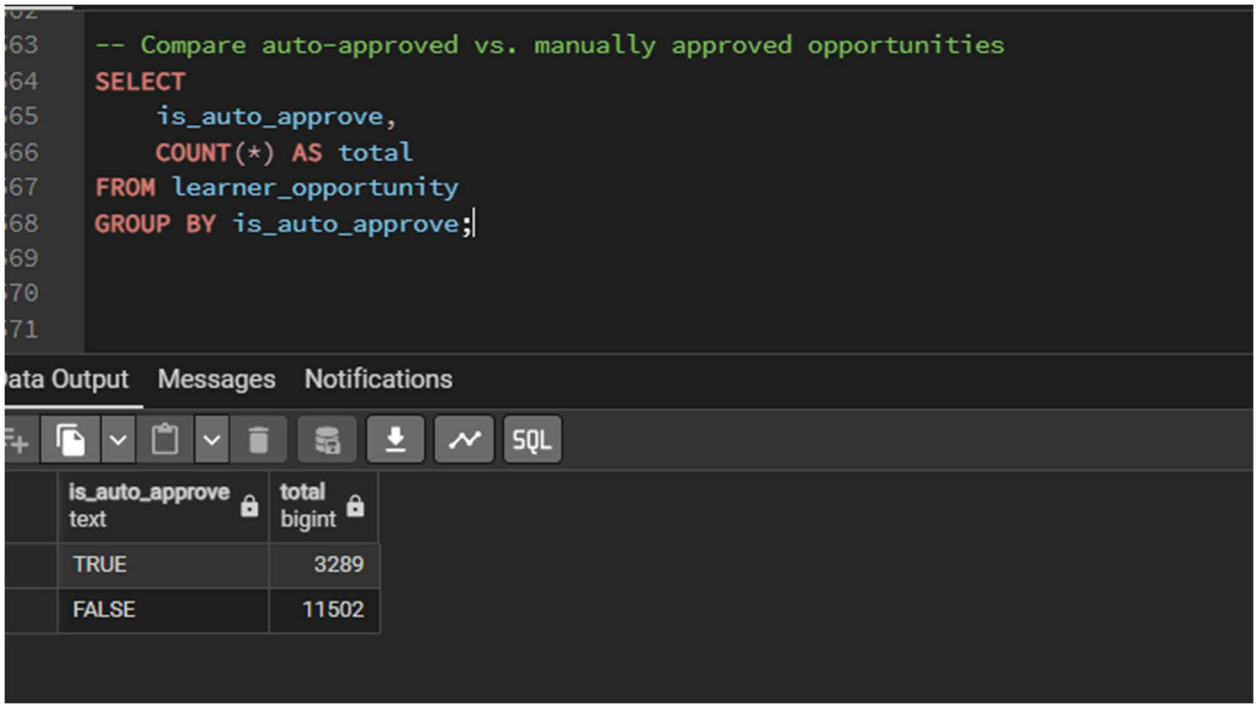


Fig 23: Auto Approval Distribution

4.11 Archived Opportunity Analysis

The archived status of opportunities was examined.

```
SELECT
   is_archived,
   COUNT(*) AS total
FROM learner_opportunity
GROUP BY is_archived;
```

Analysis:

- Distinguishes active and archived opportunities.

```
570  -- the proportion of archived opportunities
571  SELECT
572      is_archived,
573      COUNT(*) AS total
574  FROM learner_opportunity
575  GROUP BY is_archived;
576
577
578
```

Data Output | Messages | Notifications

SQL

	is_archived text	total bigint
1	TRUE	1303
2	NULL	12651
3	FALSE	837

Fig 24: Archived vs Active Opportunities

4.12 Cohort Analysis

Applications were grouped by assigned cohort.

```
SELECT
   assigned_cohort,
   COUNT(*) AS applications
FROM learner_opportunity
GROUP BY assigned_cohort
ORDER BY applications DESC
LIMIT 10;
```

Analysis:

- Identifies the cohorts with the highest participation.

```
577  -- the cohorts with the highest participation
578  SELECT
579      assigned_cohort,
580      COUNT(*) AS applications
581  FROM learner_opportunity
582  GROUP BY assigned_cohort
583  ORDER BY applications DESC
584  LIMIT 10;
585
```

Data Output | Messages | Notifications

	assigned_cohort text	applications bigint
1		9239
2	B9ZYF9D	68
3	B28EPFO	62
4	B360ZU8	32
5	BAD40TQ	31
6	BC5AWSN	30
7	B202074	22
8	B905078	20
9	BSQNZSD	19
10	BMSFPMW	18

Fig 25: Top Assigned Cohorts

4.13 Final Dataset Validation

A final validation query was executed to verify that the integrated dataset was suitable for dashboard development.

```
SELECT
    COUNT(*) FILTER (WHERE opportunity_name IS NULL) AS missing_opportunity_name,
    COUNT(*) FILTER (WHERE opportunity_category IS NULL) AS missing_category,
    COUNT(*) FILTER (WHERE payment_status IS NULL) AS missing_payment_status,
    COUNT(*) FILTER (WHERE status IS NULL) AS missing_status,
    COUNT(*) FILTER (WHERE opportunity_location IS NULL) AS missing_location
FROM learner_opportunity;
```

Validation Metric	Result
Missing Opportunity Name	1
Missing Category	7
Missing Payment Status	0
Missing Status	5,741
Missing Location	3,718

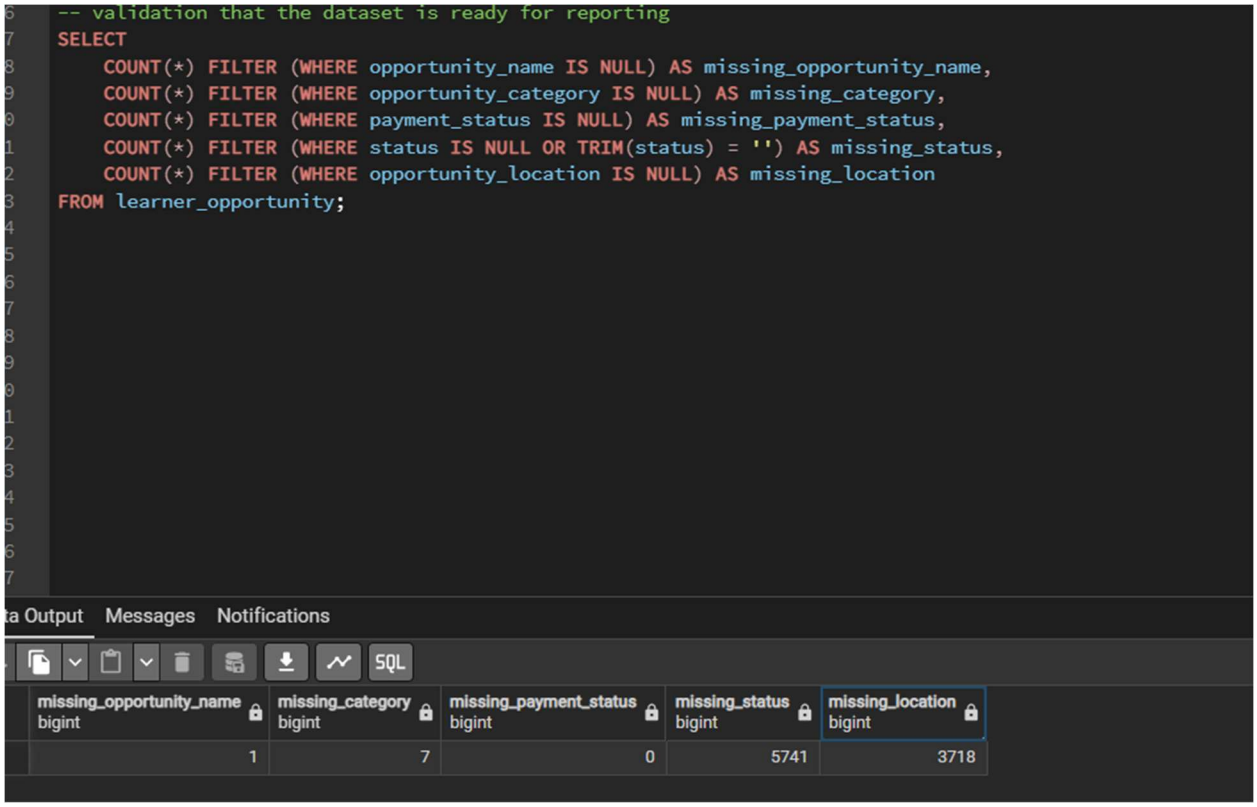


Fig 26: Final Dataset Validation Summary

The remaining missing values correspond to source-data characteristics or intentionally standardized SQL NULL values and do not affect the integrity of the integrated dataset.

Summary of the Integrated Dataset

The Opportunity and Learner datasets were successfully integrated to create a unified analytical dataset named learner_opportunity. During integration, it was observed that the learner dataset (sk) and the opportunity dataset (opportunity_id) contained the same unique identifier with different prefixes (Learner# and Opportunity#). The prefixes were normalized using the REPLACE() function, enabling a successful join between the two datasets.

The Opportunity dataset was cleaned by removing corrupted records, standardizing location values, validating categories, converting textual "NULL" values into SQL NULL, and verifying unique opportunity identifiers. The Learner dataset was cleaned by removing corrupted PDF/Base64 records, validating workflow and payment status values, reviewing date fields, and standardizing invalid category values. After integration, duplicate column names were renamed to avoid ambiguity, and blank values in important fields such as status and opportunity_location were converted to SQL NULL.

The final integrated dataset contains 14,791 matched application records and combines learner information with opportunity details, including category, location, duration, fee, eligibility, reward, workflow status, payment status, and application history. The integrated table serves as a single source of truth for analytical reporting and dashboard development.

Why Were the Remaining NULL Values Not Removed?

Although missing values were identified during validation, they were not deleted or replaced because they represent genuine missing information from the original source data rather than errors introduced during data cleaning.

The remaining missing values are:

Column	Missing Values	Reason
Opportunity Name	1	Missing in source dataset
Opportunity Category	7	Invalid `XploreU` values intentionally converted to SQL `NULL`
Payment Status	0	Fully cleaned
Status	5,741	Workflow status not available in source records; blank values standardized to SQL `NULL`
Opportunity Location	3,718	Location not provided in source records; blank values standardized to SQL `NULL`

These values were intentionally retained because replacing them with guessed or default values (such as "Unknown" or "Work From Home") would introduce inaccurate information and reduce data quality. Converting blank strings and textual "NULL" values into proper SQL NULL ensures consistent handling of missing data while preserving the integrity of the original datasets.

Dataset Readiness for Visualization

Following data integration, cleaning, and validation, the integrated dataset is ready for visualization and dashboard development.

The following validation checks confirm its readiness:

- Successfully integrated 14,791 learner application records.
- Removed corrupted and invalid records from both datasets.
- Eliminated duplicate opportunity identifiers.
- Standardized categorical values and location names.
- Converted textual "NULL" values and blank strings into SQL NULL.
- Validated workflow status and payment status values.
- Renamed duplicate column names to prevent ambiguity.
- Verified successful execution of analytical SQL queries.
- Generated summary statistics and KPI queries without errors.
- Confirmed that the remaining NULL values represent legitimate source-data limitations rather than cleaning issues.

The integrated dataset now supports a wide range of analytical operations, including KPI reporting, category analysis, payment analysis, workflow monitoring, opportunity ranking, cohort analysis, location-based analysis, and dashboard visualization in tools such as Looker Studio.

Conclusion

This report documented the complete Week 2 data preparation process, focusing on data integration, cleaning, validation, and analytical readiness. The Opportunity and Learner datasets were successfully integrated into a single dataset using normalized identifiers, resulting in 14,791 matched records.

Several data quality issues, including corrupted records, inconsistent categorical values, blank strings, and duplicate column names, were identified and resolved. Missing values originating from the source datasets were retained as SQL NULL to preserve data integrity rather than introducing assumptions through artificial replacements.

Comprehensive validation confirmed the accuracy of the integrated dataset through record count verification, duplicate checks, missing value assessment, and workflow validation. Analytical SQL queries and summary statistics further demonstrated that the dataset supports meaningful insights into applications, opportunities, payment status, workflow stages, and learner participation.

Overall, the integrated dataset is clean, consistent, validated, and ready for visualization and dashboard development. The work completed during Week 2 establishes a reliable foundation for creating interactive dashboards and performing advanced data analysis in the next phase of the internship.